

Quick Start Guide



How to Run the Library Management System

This guide will help you get the Library Management System up and running in just a few minutes.

Prerequisites

Make sure you have the following installed on your system:

- **Python 3.8+** (Download from python.org)
- **MySQL 8.0+** (Download from mysql.com)
- **Git** (for cloning the repository)

Option 1: Automated Setup (Recommended)

Step 1: Run the Setup Script

Bash

```
cd library_management_system  
./setup.sh
```

The setup script will automatically:

- Install and configure MySQL
- Create the database and tables
- Set up Python virtual environment
- Install all dependencies
- Configure environment variables

Step 2: Start the Application

Bash

```
./run.sh
```

Step 3: Access the System

Open your web browser and go to: <http://localhost:5000>

Default Login Credentials:

- Username: `admin`
- Password: `password`

Option 2: Manual Setup

Step 1: Install MySQL

Bash

```
# Ubuntu/Debian
sudo apt update && sudo apt install mysql-server

# macOS (with Homebrew)
brew install mysql

# Start MySQL service
sudo systemctl start mysql # Linux
brew services start mysql # macOS
```

Step 2: Configure MySQL

Bash

```
# Set root password
sudo mysql -e "ALTER USER 'root'@'localhost' IDENTIFIED WITH
mysql_native_password BY 'password123'; FLUSH PRIVILEGES;"

# Create database
sudo mysql < database/schema.sql
```

Step 3: Set Up Python Environment

Bash

```
cd backend/library_api

# Create virtual environment
python3 -m venv venv

# Activate virtual environment
```

```
source venv/bin/activate # Linux/macOS
# or
venv\Scripts\activate    # Windows

# Install dependencies
pip install -r requirements.txt
```

Step 4: Configure Environment

Make sure the `.env` file in `backend/library_api/` contains:

Plain Text

```
MYSQL_HOST=localhost
MYSQL_PORT=3306
MYSQL_USER=root
MYSQL_PASSWORD=password123
MYSQL_DATABASE=library_management
JWT_SECRET_KEY=your-secret-key-here
JWT_ACCESS_TOKEN_EXPIRES=3600
```

Step 5: Start the Application

Bash

```
cd backend/library_api
source venv/bin/activate
python src/main.py
```

Step 6: Access the System

Open your browser and navigate to: <http://localhost:5000>

What You Can Do

Once the system is running, you can:

As Administrator (admin/password):

-  View dashboard with library statistics
-  Add new books to the library
-  Manage book categories and details

-  View all borrowings and returns
-  Process fines and payments
-  Manage member accounts

As Regular Member:

-  Browse available books
-  Search and filter books
-  Borrow available books
-  View borrowing history
-  Renew borrowed books
-  Pay outstanding fines
-  View personal profile and statistics



Troubleshooting

MySQL Connection Issues

Bash

```
# Check if MySQL is running
sudo systemctl status mysql # Linux
brew services list | grep mysql # macOS

# Restart MySQL if needed
sudo systemctl restart mysql # Linux
brew services restart mysql # macOS
```

Python Environment Issues

Bash

```
# Make sure you're in the right directory
cd backend/library_api

# Activate virtual environment
source venv/bin/activate
```

```
# Reinstall dependencies if needed  
pip install -r requirements.txt
```

Port Already in Use

If port 5000 is already in use, you can:

1. Stop the process using port 5000
2. Or modify the port in `src/main.py`

Browser Issues

- Clear browser cache and cookies
- Try a different browser
- Check browser console for JavaScript errors



Features Overview

Dashboard

- Library statistics overview
- Recent borrowing activity
- Books due soon
- Overdue books tracking

Book Management

- Add/edit books (admin only)
- Search and filter functionality
- Category management
- Availability tracking

Member System

- User registration and login
- Profile management
- Borrowing history
- Fine tracking

Borrowing System

- One-click book borrowing
- Automatic due date calculation
- Book renewal options
- Return processing

Fine Management

- Automatic fine calculation
- Payment processing
- Fine history tracking
- Overdue notifications



User Interface

The system features a modern, responsive design that works on:

- Desktop computers
- Tablets
- Mobile phones

Key UI features:

- Beautiful gradient designs
- Smooth animations
- Interactive elements
- Toast notifications
- Modal dialogs
- Responsive tables



Security Features

- JWT token authentication
- Password hashing with bcrypt
- Role-based access control
- SQL injection prevention
- XSS protection

- Session management



Sample Data

The system comes with sample data including:

- 5 sample books across different categories
- 3 member accounts (including admin)
- Sample borrowing records
- Example fine calculations



Getting Help

If you encounter any issues:

1. **Check the logs:** Look at the Flask console output for error messages
2. **Verify prerequisites:** Ensure Python and MySQL are properly installed
3. **Check file permissions:** Make sure scripts are executable
4. **Review configuration:** Verify `.env` file settings
5. **Restart services:** Try restarting MySQL and the Flask application



Support

For additional support:

- Check the main `README.md` for detailed documentation
- Review the troubleshooting section
- Check the database schema in `database/schema.sql`
- Examine the API endpoints in the route files



Success!

Once everything is running, you should see:

- Flask server running on <http://localhost:5000>
- Beautiful login page in your browser
- Ability to log in with admin credentials
- Full access to all library management features

Enjoy using your new Library Management System! 📖✨